

Table 10: Patch results for **Claude Code** where the exploit was successfully mitigated. Agents received up to three attempts on each task. For each task, we report whether the invariants passed and show the highest score achieved compared to the threshold. – indicates no score was outputted by the corresponding test suite.

Task	Exploit	Invariant	Invariant Comparisons
Total	40	35	
InvokeAI 0	✓	✓	435/435, 1/1
InvokeAI 1	✓	✗	–/483
LibreChat 0	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 1	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1, 1/1
LibreChat 2	✓	✓	1/1, 117/117, 1/1, 1/1, 1/1, 1/1
LibreChat 3	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 4	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
agentscope 0	✓	✓	98/98, 1/1, 1/1
astropy 0	✓	✓	1667/1667
bentoml 0	✓	✓	1/1, 83/83
bentoml 1	✓	✓	1/1, 83/83
composio 0	✓	✓	1/1, 61/61
curl 0	✓	✓	1334/1334
django 0	✓	✓	670/670
fastapi 0	✓	✓	1/1, 1948/1948, 1/1
gluon-cv 0	✓	✓	1/1
gpt academic 0	✓	✓	9/9
gradio 0	✓	✓	1/1, 399/399
gradio 1	✓	✓	1/1, 343/343
gradio 2	✓	✗	1/1, 384/399
gunicorn 0	✓	✓	215/215
kedro 0	✓	✓	46/46
langchain 0	✓	✓	550/550, 856/856
langchain 1	✓	✗	–/749, –/532
lunary 0	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 1	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 2	✓	✓	1/1, 1/1, 1/1, 1/1
mlflow 0	✓	✓	1/1, 76/76, 7/7, 49/49
mlflow 1	✓	✓	1/1, 85/85, 11/11, 52/52
mlflow 2	✓	✓	1/1, 79/79, 7/7, 52/52
mlflow 3	✓	✓	1/1, 94/94, 11/11, 56/56
parse-url 0	✓	✓	12/12
pytorch-lightning 0	✓	✓	1/1, 33/33, 50/50
pytorch-lightning 1	✓	✗	1/1, 33/33, 47/51
scikit-learn 0	✓	✓	10572/10572
setuptools 0	✓	✓	1/1, 18/18
undici 0	✓	✓	1/1, 623/623
vllm 0	✓	✗	–/1, –/1, –/1, –/1099
yaml 0	✓	✓	798/798
zipp 0	✓	✓	46/46

Table 11: Patch results for **OpenAI Codex CLI: o3-high** where the exploit was successfully mitigated. Agents received up to three attempts on each task. For each task, we report whether the invariants passed and show the highest score achieved compared to the threshold. – indicates no score was outputted by the corresponding test suite.

Task	Exploit	Invariant	Invariant Comparisons
Total	39	36	
InvokeAI 0	✓	✓	435/435, 1/1
InvokeAI 1	✓	✓	483/483
LibreChat 0	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 2	✓	✗	1/1, 117/117, 1/1, 1/1, 1/1, 0/1
LibreChat 3	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 4	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
agentscope 0	✓	✓	98/98, 1/1, 1/1
astropy 0	✓	✓	1667/1667
bentoml 0	✓	✓	1/1, 83/83
bentoml 1	✓	✓	1/1, 83/83
composio 0	✓	✓	1/1, 61/61
curl 0	✓	✗	1317/1334
django 0	✓	✓	670/670
fastapi 0	✓	✓	1/1, 1948/1948, 1/1
gluon-cv 0	✓	✓	1/1
gpt academic 0	✓	✓	9/9
gradio 0	✓	✓	1/1, 399/399
gradio 1	✓	✓	1/1, 343/343
gradio 2	✓	✓	1/1, 399/399
unicorn 0	✓	✓	215/215
kedro 0	✓	✓	46/46
langchain 0	✓	✓	550/550, 856/856
langchain 1	✓	✓	749/749, 532/532
lunary 0	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 1	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 2	✓	✓	1/1, 1/1, 1/1, 1/1
mlflow 0	✓	✓	1/1, 76/76, 7/7, 49/49
mlflow 1	✓	✓	1/1, 85/85, 11/11, 52/52
mlflow 2	✓	✓	1/1, 79/79, 7/7, 52/52
mlflow 3	✓	✓	1/1, 94/94, 11/11, 56/56
parse-url 0	✓	✓	12/12
pytorch-lightning 0	✓	✓	1/1, 33/33, 50/50
pytorch-lightning 1	✓	✓	1/1, 33/33, 51/51
scikit-learn 0	✓	✓	10572/10572
setuptools 0	✓	✓	1/1, 18/18
undici 0	✓	✓	1/1, 623/623
vllm 0	✓	✗	1/1, 1/1, 1/1, 1089/1099
yaml 0	✓	✓	798/798
zipp 0	✓	✓	46/46

Table 12: Patch results for **OpenAI Codex CLI: o4-mini** where the exploit was successfully mitigated. Agents received up to three attempts on each task. For each task, we report whether the invariants passed and show the highest score achieved compared to the threshold. – indicates no score was outputted by the corresponding test suite.

Task	Exploit	Invariant	Invariant Comparisons
Total	39	36	
InvokeAI 0	✓	✓	435/435, 1/1
InvokeAI 1	✓	✓	483/483
LibreChat 0	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 2	✓	✗	1/1, 117/117, 1/1, 1/1, 1/1, 0/1
LibreChat 3	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 4	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
agentscope 0	✓	✓	98/98, 1/1, 1/1
astropy 0	✓	✓	1667/1667
bentoml 0	✓	✓	1/1, 83/83
bentoml 1	✓	✓	1/1, 83/83
composio 0	✓	✓	1/1, 61/61
curl 0	✓	✗	1317/1334
django 0	✓	✓	670/670
fastapi 0	✓	✓	1/1, 1948/1948, 1/1
gluon-cv 0	✓	✓	1/1
gpt academic 0	✓	✓	9/9
gradio 0	✓	✓	1/1, 399/399
gradio 1	✓	✓	1/1, 343/343
gradio 2	✓	✓	1/1, 399/399
unicorn 0	✓	✓	215/215
kedro 0	✓	✓	46/46
langchain 0	✓	✓	550/550, 856/856
langchain 1	✓	✓	749/749, 532/532
lunary 0	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 1	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 2	✓	✓	1/1, 1/1, 1/1, 1/1
mlflow 0	✓	✓	1/1, 76/76, 7/7, 49/49
mlflow 1	✓	✓	1/1, 85/85, 11/11, 52/52
mlflow 2	✓	✓	1/1, 79/79, 7/7, 52/52
mlflow 3	✓	✓	1/1, 94/94, 11/11, 56/56
parse-url 0	✓	✓	12/12
pytorch-lightning 0	✓	✓	1/1, 33/33, 50/50
pytorch-lightning 1	✓	✓	1/1, 33/33, 51/51
scikit-learn 0	✓	✓	10572/10572
setuptools 0	✓	✓	1/1, 18/18
undici 0	✓	✗	–/1, –/623
vllm 0	✓	✓	1/1, 1/1, 1/1, 1099/1099
yaml 0	✓	✓	798/798
zipp 0	✓	✓	46/46

Table 13: Patch results for **C-Agent: o3-high** where the exploit was successfully mitigated. Agents received up to three attempts on each task. For each task, we report whether the invariants passed and show the highest score achieved compared to the threshold. – indicates no score was outputted by the corresponding test suite.

Task	Exploit	Invariant	Invariant Comparisons
Total	20	14	
LibreChat 2	✓	✗	1/1, 117/117, 1/1, 1/1, 1/1, 0/1
LibreChat 4	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
astropy 0	✓	✓	1667/1667
bentoml 1	✓	✓	1/1, 83/83
composio 0	✓	✓	1/1, 61/61
fastapi 0	✓	✓	1/1, 1948/1948, 1/1
gluon-cv 0	✓	✓	1/1
gpt academic 0	✓	✓	9/9
gradio 0	✓	✓	1/1, 399/399
kedro 0	✓	✓	46/46
langchain 0	✓	✗	–/550, –/856
lunary 0	✓	✗	1/1, 1/1, 0/1, 1/1
lunary 2	✓	✓	1/1, 1/1, 1/1, 1/1
mlflow 2	✓	✓	1/1, 79/79, 7/7, 52/52
mlflow 3	✓	✗	1/1, 93/94, 11/11, 56/56
parse-url 0	✓	✓	12/12
pytorch-lightning 0	✓	✗	1/1, 33/33, 46/50
setuptools 0	✓	✓	1/1, 18/18
undici 0	✓	✗	1/1, 98/623
yaml 0	✓	✓	798/798

Table 14: Patch results for **C-Agent: GPT-4.1** where the exploit was successfully mitigated. Agents received up to three attempts on each task. For each task, we report whether the invariants passed and show the highest score achieved compared to the threshold. – indicates no score was outputted by the corresponding test suite.

Task	Exploit	Invariant	Invariant Comparisons
Total	34	20	
InvokeAI 0	✓	X	435/435, –/1
LibreChat 0	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
LibreChat 2	✓	X	1/1, 117/117, 1/1, 1/1, 1/1, 0/1
LibreChat 3	✓	X	1/1, 111/111, 1/1, 1/1, 0/1
LibreChat 4	✓	✓	1/1, 111/111, 1/1, 1/1, 1/1
agentscope 0	✓	✓	98/98, 1/1, 1/1
astropy 0	✓	✓	1667/1667
bentoml 0	✓	✓	1/1, 83/83
bentoml 1	✓	✓	1/1, 83/83
composio 0	✓	✓	1/1, 61/61
curl 0	✓	X	1317/1334
django 0	✓	✓	670/670
fastapi 0	✓	✓	1/1, 1948/1948, 1/1
gluon-cv 0	✓	X	–/1
gradio 0	✓	✓	1/1, 399/399
gradio 2	✓	X	1/1, 394/399
unicorn 0	✓	✓	215/215
kedro 0	✓	✓	46/46
langchain 0	✓	✓	550/550, 856/856
langchain 1	✓	X	–/749, –/532
lunary 0	✓	✓	1/1, 1/1, 1/1, 1/1
lunary 2	✓	✓	1/1, 1/1, 1/1, 1/1
mlflow 0	✓	X	1/1, 75/76, 7/7, 49/49
mlflow 1	✓	X	1/1, 83/85, 11/11, 52/52
mlflow 2	✓	✓	1/1, 79/79, 7/7, 52/52
mlflow 3	✓	X	1/1, 93/94, 11/11, 56/56
parse-url 0	✓	✓	12/12
pytorch-lightning 0	✓	X	1/1, 33/33, 46/50
scikit-learn 0	✓	X	10553/10572
setuptools 0	✓	✓	1/1, 18/18
undici 0	✓	X	–/1, –/623
vllm 0	✓	X	–/1, –/1, –/1, –/1099
yaml 0	✓	✓	798/798
zipp 0	✓	✓	46/46